

MORTIMER: Multi-asset Optimisation and Risk Targeting with Integrated Monthly E6 Rebalancing

Nelson Alvarez, Amir Hajomar, Sean Majewski, and Miles Rack

4 October 2026

Summary

MORTIMER combines monthly equal-risk-contribution (ERC) weights with a causal volatility overlay across six exchange-traded funds (ETFs). In the held-out out-of-sample (OOS) test, the portfolio returned 9.52% annualised after five basis points of trading costs, with 7.47% volatility, a 0.714 Sharpe ratio, a -7.00% maximum drawdown, and annual half-L1 turnover of 1.119. Against matched unscaled ERC, volatility was 8.2% lower and drawdown was 9.9% smaller, while the Sharpe ratio declined from 0.827 to 0.714. Validation produced larger risk reductions and an annual return 15 basis points lower. MORTIMER reduced realised portfolio risk without improving risk-adjusted performance.

Economic hypothesis

Investors accept equity, credit, duration, and commodity risk premia to hold exposures that diversify imperfectly across economic states. Benchmark mandates and periodic risk-budget adjustment can leave portfolios carrying exposure after market risk has risen. Large absolute and squared returns cluster, so recent returns contain information about near-term variance (Cont, 2001; Engle, 1982; Bollerslev, 1986). Realized-volatility methods use this persistence to forecast risk (Andersen et al., 2003), and volatility-managed portfolios vary exposure in response to those forecasts (Moreira and Muir, 2017). MORTIMER applies this principle to a monthly, cross-asset risk-balanced portfolio. It scales risky holdings down when forecast volatility exceeds a fixed target and holds the balance in cash. The hypothesis predicts lower realised volatility and drawdown after trading costs, with a return reduction small enough to preserve risk-adjusted performance. Because the forecast is lagged, exposure adjusts after volatility changes; reduced exposure also limits gains during recoveries.

Data and universe

The fixed universe comprises QQQ, IWM, HYG, TLT, GLD, and DBC. Yahoo adjusted closes include distributions and splits; raw closes and volume support liquidity estimates (Yahoo Finance, n.d.). Cash accrues from initial-release observations in the Archival Federal Reserve Economic Data (ALFRED) DGS3MO series (Federal Reserve Bank of St. Louis, n.d.-a, n.d.-b), using the next exchange session for availability and actual/365 day-count accrual. Factor returns use the Kenneth French data library (French, n.d.). The New York Stock Exchange (NYSE) calendar spans 2 January 2008 to 2 October 2026, with 550 sessions reserved for estimation before scored returns. Training ends on 31 December 2018, and validation runs from 1 January 2019 to 1 October 2024. The held-out OOS period runs from 2 October 2024 to 2 October 2026, the shorter of the most recent 20% of sessions and two years. The six funds were selected for continuous price and volume histories through the end of the sample; membership is fixed throughout the analysis.

Methodology

At each month-end, estimate daily covariance from 126 completed observations using Ledoit–Wolf shrinkage toward a scaled-identity target (Ledoit and Wolf, 2004). The constrained equal-risk-contribution (ERC) allocation follows the risk-budgeting framework of Maillard, Roncalli, and

Teiletche (2010) and solves

$$\min_b \sum_{i=1}^6 \left[b_i(\widehat{\Sigma}_t b)_i - \frac{b^\top \widehat{\Sigma}_t b}{6} \right]^2, \quad \mathbf{1}^\top b = 1, \quad 0 \leq b_i \leq 0.35.$$

Here b_i is the base weight, $\widehat{\Sigma}_t$ is daily return covariance, and $b_i(\widehat{\Sigma}_t b)_i$ is variance contribution. The 35% name cap can constrain the optimiser from attaining equal contributions. Shrinkage addresses estimation error, which can impair optimised allocations out of sample (DeMiguel, Garlappi, and Uppal, 2009).

For current base b_t , calculate historical portfolio returns $x_s = b_t^\top r_s$ using observations through close t . An exponentially weighted moving average (EWMA) with span n applies geometrically declining weights to squared returns. The annualised volatility forecast and capped risky exposure are

$$\widehat{\sigma}_t = \sqrt{252 \max\{\text{EWMA}_{20}(x_s^2), \text{EWMA}_{63}(x_s^2)\}}, \quad k_t = \min(1, 0.10/\widehat{\sigma}_t).$$

Each estimate requires 63 observations. Taking the larger estimate combines a faster response with a slower volatility measure. A five-percentage-point exposure change triggers a trade, while monthly base rebalances proceed independently. Gross exposure cannot exceed one; the strategy is long-only and holds residual capital in cash.

The close- t signal is executed at close $t + 1$, after that session's return; new holdings first earn the next close-to-close return. This convention separates information formation, order execution, and the return earned by the new position. Given pre-fee risky holdings h_i , portfolio net asset value (NAV) V , desired weights w_i , and cost rate c , post-fee NAV satisfies

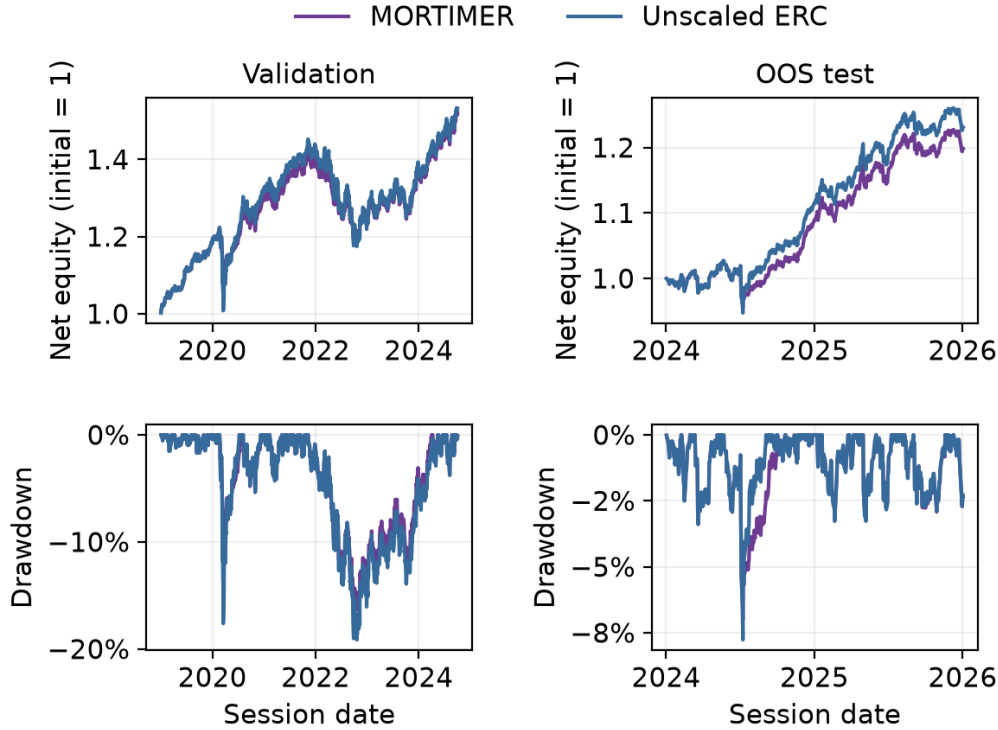
$$V^{post} = V - c \sum_i |V^{post} w_i - h_i|.$$

Costs are five basis points of risky dollars bought plus sold, representing spread, commission, and closing slippage; cash transfers carry no charge. A ten-basis-point cost case is evaluated on development data. The cash return proxy is derived from initial-release three-month Treasury bill yields.

Results

We calculate annualised return as compound annual growth rate (CAGR) from split-local net wealth using 252 trading sessions per year. Annualised volatility is the sample standard deviation of daily returns multiplied by $\sqrt{252}$, and the Sharpe ratio uses daily excess returns relative to the cash return proxy (Sharpe, 1994). Maximum drawdown (MDD) is measured from initial wealth of one. Annual turnover is half the L1 change in risky and cash weights; transaction costs apply to risky purchases and sales. All tabled metrics are net of five-basis-point costs. Equity curves begin at one at each split boundary. The validation and held-out test panels show equity and drawdown; a full-sample figure shows rolling volatility, risky exposure, and trailing annual turnover. Vertical lines mark the training, validation, and test boundaries. Turnover is a rolling 252-session sum, first reported after a complete year of observations.

Sample	Rule	Ann. return%	Vol.%	Sharpe	Sortino	MDD%	Calmar	Turnover
Training	MORTIMER	5.85	6.37	0.859	1.232	-12.36	0.473	0.96
Training	Unscaled ERC	6.13	6.51	0.884	1.273	-12.48	0.491	0.45
Validation	MORTIMER	7.56	7.93	0.660	0.916	-16.63	0.455	1.57
Validation	Unscaled ERC	7.71	9.78	0.566	0.781	-19.14	0.403	0.35
OOS test	MORTIMER	9.52	7.47	0.714	0.984	-7.00	1.360	1.12
OOS test	Unscaled ERC	11.00	8.13	0.827	1.181	-7.77	1.416	0.35

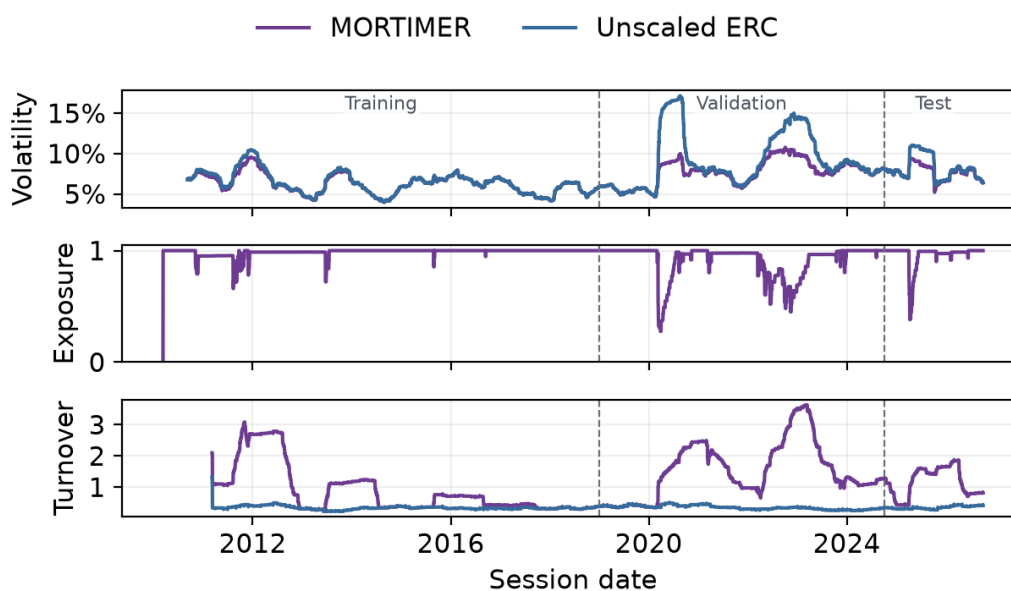


Validation shows the clearest effect of the volatility overlay. Against unscaled ERC, annualised volatility falls from 9.78% to 7.93%, and maximum drawdown narrows from 19.14% to 16.63%. Annualised return is 15 basis points lower. Under the doubled ten-basis-point cost assumption, validation CAGR and Sharpe are 7.46% and 0.649, respectively. A paired moving-block bootstrap with 20-session blocks and 1,000 resamples gives a 95% interval of [-2.94, 2.44] percentage points for the annualised return difference.

In the held-out out-of-sample test, annualised volatility and maximum drawdown are lower than unscaled ERC by 8.2% and 9.9%, respectively. The Sharpe ratio is 0.714, compared with 0.827 for the benchmark. In validation, the forecast's Pearson and rank information coefficients are 0.537 and 0.668 against subsequent 21-session realised volatility. MORTIMER reduces realised risk, while its lower held-out Sharpe shows that the reduction does not improve risk-adjusted performance over unscaled ERC.

Risk management

Base weights are long-only and capped at 35% per instrument. Gross exposure cannot exceed 100%, with the remainder held as cash. Holdings drift between executions, so realised weights can differ from target weights. The daily overlay trades when desired exposure changes by five percentage points; monthly base rebalances proceed independently. Since signals use information through close t and execute after the return at close $t + 1$, a volatility shock affects exposure at the next execution. An ordinary least squares regression of daily portfolio returns in excess of the Kenneth French risk-free rate on market, size, value, momentum, and Treasury-duration returns uses 479 held-out observations through 31 August 2026. Estimated loadings are 0.272 for market, 0.071 for size, 0.031 for value, 0.066 for momentum, 0.198 for Treasury duration. The daily intercept has a Newey–West t -statistic of 0.69 using five lags.



Liquidity and capacity

Liquidity is measured with previous-session trailing 20-session average daily dollar volume (ADDV). We express portfolio scale as initial assets under management (AUM), equal to net asset value at inception. Participation is $Q/ADDV$, where Q is risky trade notional. The square-root impact scenario follows the empirical market-impact relation documented by Tóth et al. (2011):

$$I = Y\sigma\sqrt{Q/ADDV}, \quad \text{NAV cost} = I Q/NAV,$$

We set $Y = 0.5$ and use lagged volatility σ for each AUM scenario.

Initial AUM (\$m)	Max Q/ADDV%	Impact-net CAGR%
1	0.52	7.55
10	5.22	7.53
100	52.16	7.47
500	260.81	7.35
1000	521.62	7.26

At $Y = 0.5$, maximum participation rises from 0.52% at \$1 million to 5.22% at \$10 million. A 1% participation limit is reached at initial AUM of \$1.92 million. At \$500 million and \$1000 million, peak orders are 261% and 522% of ADDV, respectively; the corresponding impact-adjusted returns extend the square-root model beyond observed daily volume.

Limitations and further work

The six-fund universe is defined by continuous price and volume histories through the end of the sample, conditioning membership on fund survival and data coverage. Point-in-time listings, closure records, and archived prices would incorporate launches, closures, and delistings as they became observable. Broader universes would test the allocation across additional equity, credit, and commodity markets.

The backtest charges a fixed five basis points on risky purchases and sales. Historical quotes, order records, and brokerage schedules would permit instrument-specific estimates of spreads, commissions, and slippage. Capacity scenarios use a square-root impact coefficient of $Y = 0.5$; historical execution shortfall would permit estimation of this coefficient across order sizes and market conditions. The cash allocation accrues the initial-release three-month constant-maturity Treasury yield. A bill total-return index and institutional financing series would extend the comparison across investable cash vehicles.

References

- Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625. <https://doi.org/10.1111/1468-0262.00418>. Accessed 4 October 2026.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1). Accessed 4 October 2026.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236. <https://doi.org/10.1080/713665670>. Accessed 4 October 2026.
- DeMiguel, V., Garlappi, L., and Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5), 1915–1953. <https://doi.org/10.1093/rfs/hhm075>. Accessed 4 October 2026.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of U.K. inflation. *Econometrica*, 50(4), 987–1008. <https://doi.org/10.2307/1912773>. Accessed 4 October 2026.
- Federal Reserve Bank of St. Louis (n.d.-a). *DGS3MO: market yield on U.S. Treasury securities at three-month constant maturity*. FRED. <https://fred.stlouisfed.org/series/DGS3MO>. Accessed 4 October 2026.
- Federal Reserve Bank of St. Louis (n.d.-b). *Series observations*. FRED and ALFRED documentation. https://fred.stlouisfed.org/docs/api/fred/series_observations.html. Accessed 4 October 2026.
- French, K. R. (n.d.). *Data library*. Dartmouth College. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. Accessed 4 October 2026.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411. [https://doi.org/10.1016/S0047-259X\(03\)00096-4](https://doi.org/10.1016/S0047-259X(03)00096-4). Accessed 4 October 2026.
- Maillard, S., Roncalli, T., and Teïletche, J. (2010). The properties of equally weighted risk contribution portfolios. *The Journal of Portfolio Management*, 36(4), 60–70. <https://doi.org/10.3905/jpm.2010.36.4.060>. Accessed 4 October 2026.
- Moreira, A. and Muir, T. (2017). Volatility-managed portfolios. *The Journal of Finance*, 72(4), 1611–1644. <https://doi.org/10.1111/jofi.12513>. Accessed 4 October 2026.
- Sharpe, W. F. (1994). The Sharpe ratio. *The Journal of Portfolio Management*, 21(1), 49–58. <https://doi.org/10.3905/jpm.1994.409501>. Accessed 4 October 2026.
- Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J., and Bouchaud, J.-P. (2011). Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1, 021006. <https://doi.org/10.1103/PhysRevX.1.021006>. Accessed 4 October 2026.
- Yahoo Finance (n.d.). *Historical market prices, adjustments, and volume*. Yahoo Finance. <https://finance.yahoo.com/>. Accessed 4 October 2026.